Lecture-24

21/11/2025

Internal direct Sum

Let W_1 and W_2 be two Subspaces of a Vector Space V.
We Say that the sum

$$W_1 + W_2 = \{ \omega_1 + \omega_2 : \omega_1 \in W_1, \ \omega_2 \in W_2 \}$$ is an

internal direct Sum if $\omega_1 + \omega_2 = 0$ for $\omega_1 \in W_1$ and $\omega_2 \in W_2$ implies $\omega_1 = 0$ and $\omega_2 = 0$.

We will use the notation $W_1 \oplus W_2$ to denote $W_1 + W_2$ when the latter is an internal direct sum of W_1 and W_2.

Examples Let $W_1 = \left\{ \begin{pmatrix} x \\ 0 \end{pmatrix} : x \in \mathbb{R} \right\}$ and $W_2 = \left\{ \begin{pmatrix} 0 \\ y \end{pmatrix} : y \in \mathbb{R} \right\}$ be Subspace of $\mathbb{R}^2$. If $\omega_1 \in W_1$ and $\omega_2 \in W_2$,

Say $\omega_1 = \begin{pmatrix} x \\ 0 \end{pmatrix}$ and $\omega_2 = \begin{pmatrix} 0 \\ y \end{pmatrix}$, then

$$\omega_1 + \omega_2 = 0 \Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow x = 0 \text{ and } y = 0.$$

That is, $\omega_1 = 0$ and $\omega_2 = 0$. Thus $W_1 + W_2$ is an internal direct Sum of W_1 and W_2.

② Let $W_1 = \left\{ \begin{pmatrix} x \\ y \\ 0 \end{pmatrix} : x, y \in \mathbb{R} \right\}$ and $W_2 = \left\{ \begin{pmatrix} x \\ 0 \\ z \end{pmatrix} : x, z \in \mathbb{R} \right\}$ be Subspace of $\mathbb{R}^3$. Then, we have $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \in W_1$ and $\begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix} \in W_2$ and that $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix} = 0$. Therefore $W_1 + W_2$ is not an internal direct Sum of Subspaces W_1 and W_2.

Proposition

Let W_1 and W_2 be two subspaces of a vector space V. Then, $W_1 + W_2$ is an internal direct sum if and only if $W_1 \cap W_2 = \{0\}$

Proof

$\Longrightarrow$ Suppose that $W_1 + W_2 = W_1 \oplus W_2$. If $w \in W_1 \cap W_2$ then since $w + -w = 0$ and $w \in W_1$ and $-w \in W_2$, we must have $w = 0$. Hence $W_1 \cap W_2 = \{0\}$

$\Longleftarrow$ Suppose that $W_1 \cap W_2 = \{0\}$. Let $w_1 \in W_1$ and $w_2 \in W_2$ be elements such that $w_1 + w_2 = 0$

Then, $w_1 = -w_2$ and we obtain that $w_1 \in W_2$.

That is, $w_1 \in W_1 \cap W_2 = \{0\}$. This implies $w_1 = 0$ and consequently, $w_2 = 0$ $\square$

Direct Sum of finitely many Subspaces

Let $W_1, W_2, \ldots, W_n$ be finitely many subspaces of an F-vector space V. Then

$$W_1 + W_2 + \cdots + W_n = \{ w_1 + w_2 + \cdots + w_n : w_i \in W_i \text{ for each } i \}$$

is a direct sum of subspaces if $w_1 + w_2 + \cdots + w_n = 0$ for $w_1 \in W_1, \ldots, w_n \in W_n$ implies $w_1 = 0, w_2 = 0, \ldots, w_n = 0$.

Example Let $\{e_1, \cdots, e_n\}$ be the standard basis of $\mathbb{R}^n$.
Let $W_1 = \text{Span}\{e_1\}, \ldots, W_n = \text{Span}\{e_n\}$. Then the sum

$$\sum_{i=1}^{n} W_i = \bigoplus_{i=1}^{n} W_i$$

For $\omega_1 \in W_1, \ldots, \omega_n \in W_n$,

$$\omega_1 + \omega_2 + \cdots + \omega_n = 0$$

$$\Rightarrow \alpha_1 e_1 + \alpha_2 e_2 + \cdots + \alpha_n e_n = 0$$

$$\Rightarrow \alpha_1 = 0, \alpha_2 = 0, \ldots, \alpha_n = 0$$

$$\Rightarrow \omega_1 = 0, \omega_2 = 0, \ldots, \omega_n = 0$$

Proof: Easy!

Proposition Let $W_1, W_2, \cdots, W_m$ be subspaces of a finite dimensional vector space V. Suppose that $W_1 + \cdots + W_n = W_1 \oplus \cdots \oplus W_n$.
Then, $\text{Dim}_F(W_1 \oplus \cdots \oplus W_n) = \text{Dim}_F(W_1) + \text{Dim}_F(W_2) + \cdots + \text{Dim}_F(W_n)$

Proof: Let $m = 2$. Then since

$$\text{Dim}_F(W_1 + W_2) = \text{Dim}_F(W_1) + \text{Dim}_F(W_2) - \text{Dim}_F(W_1 \cap W_2)$$

and $W_1 \cap W_2 = 0$, we have

$$\text{Dim}_F(W_1 + W_2) = \text{Dim}_F(W_1) + \text{Dim}_F(W_2)$$

Assume that for any $m-1$ subspaces the Proposition holds. (induction)

Now,

$$\text{Dim}_F\left(\sum_{i=1}^{m} W_i\right) = \text{Dim}_F\left(\sum_{j=1}^{m-1} W_j\right) + \text{Dim}_F(W_m) - \text{Dim}_F\left(\sum_{j=1}^{m-1} W_j \cap W_m\right)$$

Induction hypothesis

$$= \sum_{j=1}^{m-1} \text{Dim}_F(W_j) + \text{Dim}_F(W_m) - \text{Dim}_F\left(\sum_{j=1}^{m-1} W_j \cap W_m\right)$$

$$= \sum_{i=1}^{m} \mathrm{Dim}_F(W_j) - \mathrm{Dim}_F\left(\sum_{j=1}^{m-1} W_j \cap W_m\right) \quad \text{Ⓐ}$$

Thus, it is enough to show that $\displaystyle\sum_{j=1}^{m-1} W_j \cap W_m = 0$.

Let $\omega \in \displaystyle\sum_{j=1}^{m-1} W_j \cap W_m$. Then $\omega = \omega_1 + \omega_2 + \cdots + \omega_{m-1}$, where $\omega \in W_m$, $\omega_1 \in W_1, \ldots, \omega_{m-1} \in W_{m-1}$.

Same as $m=2$ Case

Since

$$\sum_{i=1}^{m} W_i = \bigoplus_{i=1}^{m} W_i, \qquad \omega_1 + \omega_2 + \cdots + \omega_{m-1} + -\omega = 0$$

$$\Rightarrow \quad \omega_1 = 0,\ \omega_2 = 0,\ \ldots,\ \omega_{m-1} = 0,\ \omega = 0.$$

Now from Ⓐ, we obtain that

$$\mathrm{Dim}_F(W_1 \oplus \cdots \oplus W_n) = \mathrm{Dim}_F(W_1) + \mathrm{Dim}_F(W_2) + \cdots + \mathrm{Dim}_F(W_n) \qquad \square$$

Theorem Let $T: V \longrightarrow W$ be a linear transformation of finite dimensional vector spaces. There is a subspace U of V such that U is isomorphic to $\mathrm{im}(T)$ and

$$V = \ker(T) \oplus U$$

Proof Let $\{v_1, \ldots, v_\ell\}$ be a basis of $\ker(T)$. Extend this linearly independent set to a basis $\{v_1, \ldots, v_\ell, v_{\ell+1}, \ldots, v_n\}$ of V. As in the proof of rank-nullity theorem, the set $\{T(v_{\ell+1}), \ldots, T(v_n)\}$

is linearly independent and it spans $\mathrm{im}(T)$.

Let $U = \mathrm{Span}\{v_{\ell+1}, v_{\ell+2}, \dots, v_n\}$. Then $V = \ker T + U$ as $V = \mathrm{Span}\{v_1, \dots, v_\ell, v_{\ell+1}, \dots, v_n\}$. If $v \in \ker T \cap U$ then

$$v = \alpha_1 v_1 + \dots + \alpha_\ell v_\ell = \beta_{\ell+1} v_{\ell+1} + \dots + \beta_n v_n .$$

This implies, $\alpha_1 v_1 + \dots + \alpha_\ell v_\ell - \beta_{\ell+1} v_{\ell+1} - \dots - \beta_n v_n = 0$.

Therefore, $\alpha_1 = 0, \alpha_2 = 0, \dots, \alpha_\ell = 0, \beta_{\ell+1} = 0, \dots, \beta_n = 0$.

Hence $v = 0$ and we obtain

$$V = \ker T \oplus U .$$

Finally, $\mathrm{im}\,T = \mathrm{Span}\{T(v_{\ell+1}), \dots, T(v_n)\}$ and

$U = \mathrm{Span}\{v_{\ell+1}, \dots, v_n\}$ bases!

and the map

$$\varphi : U \longrightarrow \mathrm{im}(T)$$

defined by $\varphi\left(\sum_{j=\ell+1}^{n} \alpha_j v_j\right) = \sum_{j=\ell+1}^{n} \alpha_j T(v_j)$ is

easily seen to be an isomorphism of Vector Spaces $\square$

H.W: Any two finite dimensional Vector Spaces V and W with $\mathrm{Dim}_F(V) = \mathrm{Dim}_F(W)$ are isomorphic.

Hint: If $n = \mathrm{Dim}_F(V) = \mathrm{Dim}_F(W)$ then

$$V \cong F^n \text{ and } W \cong F^n$$